

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

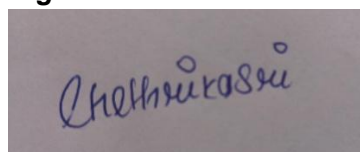
CONCEPT MAPPING

Code of Conduct:

I _____chethrikasri_(192511112)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'Chethrikasri' written in a cursive style.

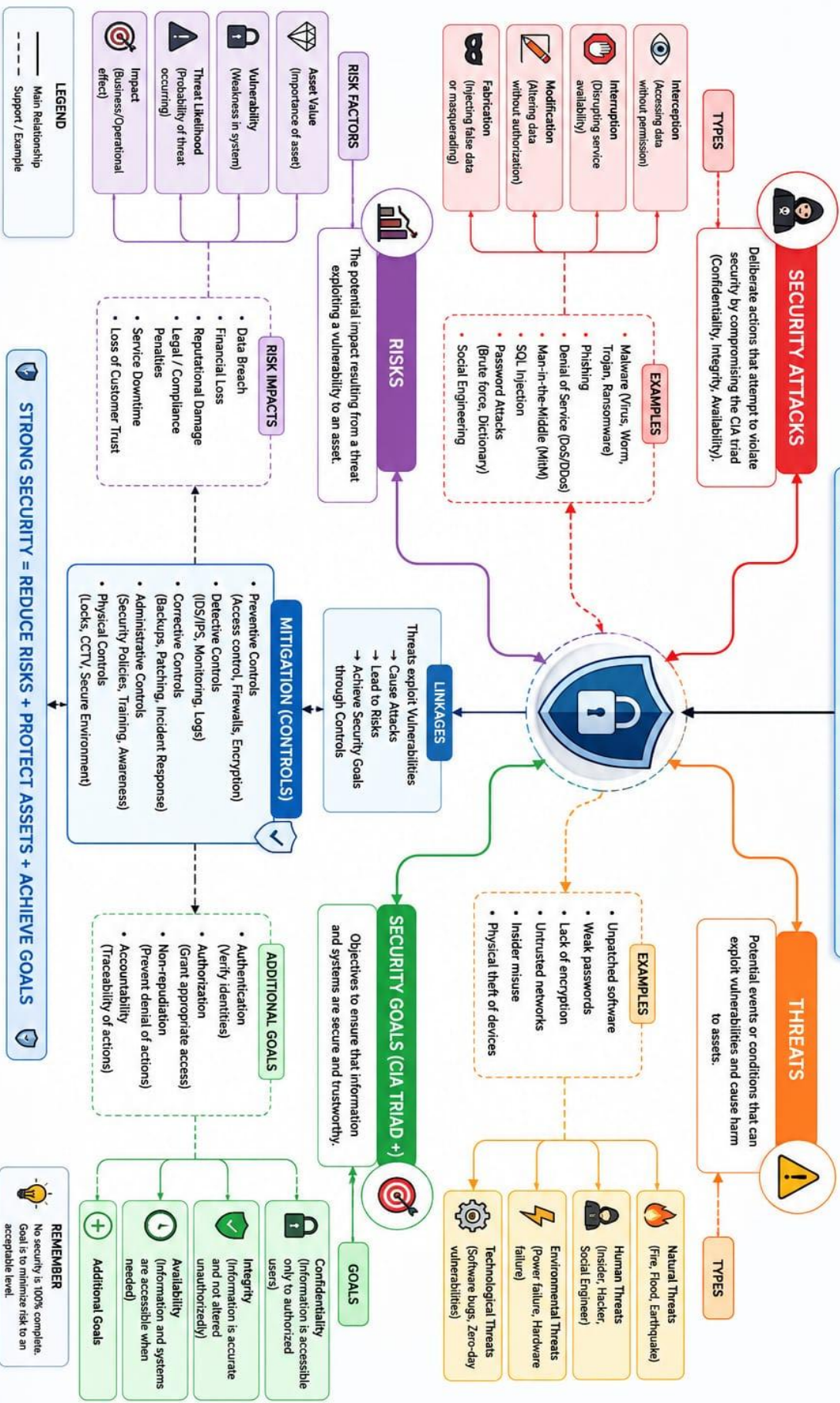
Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

SECURITY ATTACKS, THREATS, RISKS

Protecting information and systems from unauthorized access, damage, disruption or disclosure while ensuring safe and reliable use.



1 CAESAR CIPHER

Each letter is shifted by a fixed number (k) through the alphabet.

KEY USAGE
Single integer (shift value k)
Example: k = 3

ENCRYPTION RULE
 $C = (P + k) \bmod 26$

DECRYPTION RULE
 $P = (C - k) \bmod 26$

EXAMPLE (k = 3)
Plaintext : HELLO
Ciphertext: KHOOR

COMPLEXITY
Very Low
(Only 25 possible keys)

SECURITY STRENGTH
Very Weak
(Easily broken by brute force)

2 MONOALPHABETIC CIPHER

Each letter is replaced by another letter using a fixed substitution alphabet.

KEY USAGE
A 26-letter permutation key
(Example: QWERTYUIOPASDFGHJKLZXCVBNM)

ENCRYPTION
Each plaintext letter is replaced by the corresponding letter in the key.

DECRYPTION
Use inverse mapping of the key.

EXAMPLE
Key: QWERTYUIOPASDFGHJKLZXCVBNM
Plaintext : CRYPTO
Ciphertext: KFYVQA

COMPLEXITY
Medium
(26! possible keys $\approx 4.03 \times 10^{26}$)

SECURITY STRENGTH
Weak to Moderate
(Frequency analysis can break it)

KEY INSIGHTS

- All substitution ciphers replace letters; they differ in key usage and complexity.
- As complexity increases, security generally improves.
- Classical ciphers form the foundation of modern cryptography.

CONCEPT MAP – SUBSTITUTION TECHNIQUES

Caesar Cipher • Monoalphabetic Cipher • Vigenère Cipher • Hill Cipher

Substitution ciphers replace plaintext letters (or blocks) with other letters (or blocks) according to a rule using a key.



3 VIGENÈRE CIPHER

Uses a keyword to determine the shift for each letter.

KEY USAGE
A keyword of any length
(Example: KEY)

ENCRYPTION RULE
 $C_i = (P_i + K_i) \bmod 26$
(Where K_i is the numerical value of the i^{th} key letter)

DECRYPTION RULE
 $P_i = (C_i - K_i) \bmod 26$
(Repeat the keyword over the plaintext)

EXAMPLE (Key = KEY)
Plaintext : ATTACKATDAWN
Key : K E Y K E Y K E Y K
Ciphertext: KXRGIXXBVYNK

COMPLEXITY
High for long keys
(Key space = 26^n , n = key length)

SECURITY STRENGTH
Moderate to Strong
(Resistant to simple frequency analysis)

Key Usage

Encryption / Rule

Example

Complexity

Decryption / Rule

Security

LEGEND

4 HILL CIPHER

Encrypts blocks of letters using matrix multiplication.



KEY USAGE
Invertible matrix (n x n) modulo 26
(Example 2 x 2 key matrix)

ENCRYPTION RULE
 $C = K \times P \pmod{26}$
(P and C are vectors of size n)

DECRYPTION RULE
 $P = K^{-1} \times C \pmod{26}$
(Where K^{-1} is the inverse of K modulo 26)

EXAMPLE (Using 2x2 Key)
Key $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$
Plaintext (letters): HE AT $\rightarrow$ [7 4] [0 19]
Ciphertext : HI HI KF

COMPLEXITY
High
(Matrix operations, requires inversion)

SECURITY STRENGTH
Strong for small n, increases with n
(Vulnerable to known-plaintext attacks for small n)



COMPARISON SUMMARY

Cipher	Key Usage	Complexity	Security Strength	Best For
Caesar	Single shift (integer)	Very Low	Very Weak	Learning & Simple use
Monoalphabetic	26-letter substitution	Medium	Weak to Moderate	Basic substitution concepts
Vigenère	Keyword (any length)	High	Moderate to Strong	Better security with longer keys
Hill	Matrix (n x n) mod 26	High	Strong (small n) Stronger with larger n	High security applications (block based)

CONCLUSION

Among the five, Hill Cipher offers stronger security for short messages (block based), while Vigenère Cipher is preferred for longer messages with strong keywords.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation